Test Plan for Automation Exercise Web Application

Project Name: Automation Exercise E-commerce Testing Project
Test Plan ID: TP-PB-001
Prepared By: Yasmin Khaled Ahmed
Date: *April 2026*
Tested URL: https://automationexercise.com/

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1. Introduction

The purpose of this test plan is to define the testing strategy, scope, objectives, resources, and schedule for validating the Automation Exercise e-commerce web application.

Automation Exercise simulates a real online shopping system where users can register, login, search products, add items to cart, and complete checkout processes.

The project also includes API testing using the public API endpoints of the system and database testing using the Sakila sample database to practice SQL validation techniques.

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2. Objectives

- Verify that the main website features (registration, login, product search, cart, and checkout) function correctly through manual and automation testing.
- Validate API endpoints by checking response status codes, data accuracy, and proper handling of valid and invalid requests.
- Perform database testing using SQL queries to verify data retrieval, relationships between tables, and correctness of query results.
- Identify defects and ensure the system meets functional requirements with proper input validation and expected behavior.

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3. Scope

In Scope:

Website Testing (Automation Exercise)

- User registration
- User login and logout.

- Product browsing.
- Product search functionality.
- Add to cart and remove from cart.
- Checkout process.
- Contact us form.
- Subscription form.
- API requests and responses.
- UI elements validation.
- Navigation links.
- Form validation.
- Error messages.
- Forget Password.
- Update profile information.

Database Testing (Sakila Database)

- Verify data retrieval using SELECT queries
- Test relationships between tables
- Practice JOIN queries
- Validate data integrity
- Verify correct results returned from database queries

Out of Scope:

- Payment gateway integration
- Performance testing
- Backend database testing.
- Security penetration testing.
- Load testing.

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4. Test Items / Features to be Tested

A. Website Modules (Automation Exercise)

Module	Features to Test
Registration	valid inputs, invalid inputs, required fields
Login	Valid/invalid credentials, password case sensitivity, session handling
Products	product list display
Search	search results accuracy
Cart	add/remove items
Checkout	order process steps
Contact Us	form validation
Subscription	email validation
Navigation	menu links
Logout	session termination, , redirect to homepage

B. API Testing Modules

API Feature	Test Focus
Get products list	response status and data
Create user	successful creation
Verify login	authentication validation
Delete user	successful deletion

C. Database Testing Modules (Sakila)

Table	Test Scenario
actor	retrieve actor details
film	retrieve film list
film_actor	validate relationship
customer	retrieve customer data
rental	validate rental records

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5. Test Approach

Testing Types:

- **Manual Testing**
- **Automation Testing (Selenium)**
- **API Testing (Postman)**
- **Database Testing (SQL queries)**
- **Functional Testing**
- **UI/UX Testing**
- **Input Validation Testing**
- **Exploratory Testing**

Techniques:

- Black-box testing.
- Positive and negative test scenarios.
- Boundary value analysis
- Equivalence partitioning.

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6. Test Environment

Item	Description
Environment	Web-based application
URL	https://automationexercise.com/
Database	Sakila sample database
Browser	Chrome (latest), Firefox, Edge
OS	Windows 10/11
Tools	Selenium, Postman, MySQL Workbench, Excel

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7. Test Data

Example Test Data for Registration:

Field	Valid Input	Invalid Input
Name	Shahd	"" (blank)
Address	Cairo	@@@@
State	Egypt	(blank)
phone	0123456789	Abc
Email	test@gmail.com	Test
Password	Test@123	123

Example SQL Queries (Sakila)

- Retrieve actors:

```
SELECT * FROM actor;
```

- Join film and actor tables:

```
SELECT film.title, actor.first_name
FROM film
JOIN film_actor ON film.film_id = film_actor.film_id
JOIN actor ON actor.actor_id = film_actor.actor_id;
```

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8. Entry and Exit Criteria

Entry Criteria:

- Website is deployed and accessible.
- Test tools installed
- Test data prepared
- Database installed locally

Exit Criteria:

- All test cases executed.
- All critical and major defects resolved.
- Test summary report prepared.

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9. Deliverables

- Test Plan Document
- Test Cases Document
- Defect Log
- Test Execution Report
- Automation Test Scripts (Selenium)
- API Test Collection (Postman)
- SQL Queries File (Database Testing)
- Test Summary Report

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10. Risks and Mitigation

Risk	**Mitigation**
Website downtime (demo server)	Retry after some time
test data reset	recreate test account
API request failure	retry request
database errors	recheck SQL query

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11. Test Schedule (Not Final)

Activity	Start Date	End Date
Test Plan Preparation	02 Apr 2026	04 APR 2026
Test Case Design	05 Apr 2026	25 Apr 2026
Manual Testing	26 Apr 2026	17 May 2026
API + Database Testing	18 May 2026	08 Jun 2026
Automation Testing	09 Jun 2026	30 Jun 2026
Test Closure	01 Jul 2026	10 Jul 2026

Team Members

Yasmin Khaled Ahmed	**Team Leader/QC**
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Mina Adel Kadess	**QC Engineer**
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